

TB Inverted Flanger

DETAILED ENGLISH USER MANUAL

A phase-inverted short-delay modulation effect for hollow cancellation, metallic sweeps, and focused sci-fi movement.



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Host-visible endpoints documented: 7

1. Product purpose

A phase-inverted short-delay modulation effect for hollow cancellation, metallic sweeps, and focused sci-fi movement.

Traditional flanging adds a modulated delay copy to the dry signal. TB Inverted Flanger reverses the phase relationship of the delayed voices, creating deeper cancellation and a more empty, subtractive character while retaining the recognizable sweep of a flanger.

What result it creates

- Hollow phase-cancelled movement
- Metallic jet and comb sweeps
- Subtle animated width at low mix
- Repeatable pedal-style modulation with six programs

Signal flow

Dry input -> shared LFO -> three short inverted delay voices -> feedback loop -> Color core balance -> final Mix

2. Complete feature guide

Depth

Sets how far delay time moves. More depth increases sweep width and makes the flanger identity more obvious.

Rate

Controls LFO speed from very slow motion to a 10 Hz scanner-like modulation.

Feedback

Returns the inverted delayed signal to the delay path, increasing comb resonance and metallic intensity.

Color

Controls extra dry body inside the effect core. Higher Color emphasizes the hollow inverted-delay character by reducing that body.

Mix

Blends the original signal with the complete flanger core. Use low values for motion around an intact source and high values for exposed cancellation.

Programs

Empty Orbit, Null Corridor, Neon Comb, Signal Rift, Slow Eclipse, and 10Hz Scanner demonstrate the full movement range.

3. Quick start

1. Start from Empty Orbit.
2. Set Mix high enough to hear the effect clearly.

3. Match Rate to the phrase or tempo.
4. Increase Depth for a wider sweep.
5. Raise Feedback carefully for resonance.
6. Use Color to decide how hollow the center should become.

4. Practical workflows

Sci-fi vocal

1. Use medium Rate and high Color.
2. Increase Feedback until metallic formants appear.
3. Blend Mix below full wet to preserve intelligibility.

Expected result: The voice gains moving hollow sidebands while words remain recognizable.

Slow synth orbit

1. Choose a very low Rate.
2. Use high Depth and moderate Feedback.
3. Automate Mix into transitions.

Expected result: Sustained harmonics move through wide subtractive notches.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Cancellation depends on the source and can reduce low frequencies unexpectedly.
- High Feedback can become sharp and loud.
- Saved older instances may recall older defaults; use a new instance when comparing factory behavior.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.

- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 7 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Color VST3 ID 94842723	0.0 to 100.0	68.0	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Depth VST3 ID 95472323	0.0 to 100.0	72.0	Automatable	Sets how strongly the named process affects the signal.
Feedback VST3 ID 1955982213	0.0 to 100.0	32.0	Automatable	Returns processed signal to its own input, extending or intensifying the effect.
Mix VST3 ID 108124	0.0 to 100.0	52.0	Automatable	Sets the balance between the original signal and the complete processed path.
Program VST3 ID 1886548852	Empty Orbit, Null Corridor, Neon Comb, Signal Rift, Slow Eclipse, 10Hz Scanner	Empty Orbit	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Rate VST3 ID 3493088	0.030 to 10.000	0.340	Automatable	Sets the speed of modulation, movement, or repeated change.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.